Edge-Enhanced Deep Super-Resolution for Thermal Imagery

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Abstract

Infrared imagery is crucial for surveillance, object detection, and autonomous navigation applications. However, the inherent low resolution of IR sensors limits the extraction of fine-grained details necessary for downstream tasks. Previous super-resolution approaches have predominantly optimized performance for IR single-image super-resolution; however, existing methods overlook spatial error analysis and real-world validation. This article presents an Edge Enhanced Deep Super-Resolution Network specifically designed for IR images by integrating a three-component loss function; pixel-wise L1 loss, edge-preserving loss, and perceptual loss. EEDSR is specifically tailored to ensure color fidelity, sharp edge reconstruction, and perceptually realistic outputs, unlike prior methods. Experiments performed on benchmark datasets and real-world captured IR images demonstrate significant improvements over existing methods, achieving a PSNR of 30.47 and an SSIM of 0.8810. Furthermore, comprehensive error map analysis highlights superior structural preservation and reduced spatial artifacts. The proposed approach bridges the gap between high-fidelity IR reconstruction and practical applicability, paving the way for more vigorous IR-based vision systems. Suárez, Patricia L., Dario Carpio, and Angel D. Sappa. "Enhancement of Guided Thermal Image Super-Resolution Approaches." *Neurocomputing* 573 (March 2024): 127197. https://doi.org/10.1016/j.neucom.2023.127197.

CCS Concepts

• **Computing methodologies**;

Keywords

Infrared Super-Resolution, Edge Enhancement, Error Map Analysis, M3FD Dataset, Real-World Validation, Thermal Imaging, Deep Learning, EEDSR

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1 Introduction

Single Image Super-Resolution reconstructs high-resolution (HR) images from low resolution (LR) counterparts and is an ill-posed problem central to computer vision [Yang et al., 2010]. While deep learning has dramatically advanced SISR for RGB imagery [Dong et al., 2014, Huang et al., 2025] its application to infrared (IR) thermal imaging presents unique challenges due to the distinct physical characteristics of thermal radiation capture [Che et al., 2025, Kim et al., 2016].

Thermal SISR enables critical applications including surveillance, autonomous navigation, industrial inspection, and medical thermography [Huang et al., 2021, Lim et al., 2017], where preserving edges and thermal consistency directly impacts downstream tasks like object detection and anomaly recognition. Early CNN-based approaches learned direct LR-to-HR mappings [Dong et al., 2014, Zhang et al., 2018], while residual networks improved fidelity by learning intensity residuals [Wang et al., 2018]. However, these methods often blur thermal edges and lack explicit mechanisms to preserve structural boundaries [Liu et al., 2022].

Recent transformer-based models like SwinIR [Liang et al., 2021], excel in RGB super-resolution through self-attention, but our experiments reveal significant thermal domain limitations. On M3FD [Huang et al., 2024], SwinIR achieves competitive PSNR (30.89 dB) and SSIM (0.9314) but introduces sharpening artifacts and fails to preserve thermal consistency. On real-world IR data, performance drops sharply (25.60 dB PSNR, 0.7670 SSIM), underscoring poor generalization to infrared-specific degradations [Jiang et al., 2023].

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Most IR SISR methods rely on bicubic degradation models [Li et al., 2025, Suárez et al., 2024], lack real-world validation, and optimize primarily for pixel-level metrics, neglecting spatial error analysis and edge preservation. To address these gaps, we propose **Edge-Enhanced Deep Super Resolution** a CNN based framework integrating explicit edge guidance, error-aware refinement, and multi-component loss (L1 + edge + perceptual) tailored for thermal imagery. EEDSR bridges the simulation-to-reality gap, improving both metric scores and visual fidelity across benchmark and real captured datasets.

2 Proposed Method

We propose the Edge-Enhanced Deep Super-Resolution (EEDSR) network to address critical edge-preservation [Johnson et al., 2016] limitations in infrared super-resolution. Building on EDSR [Lim et al., 2017], EEDSR introduces explicit edge guidance and error-aware refinement through a multi-component loss function.

2.1 Architecture

As shown in Figure 1, EEDSR processes low-resolution infrared input through parallel pathways: (1) a main stream using residual blocks, and (2) an edge guidance branch extracting structural information via Sobel filters. Features are concatenated, refined error-aware, and up sampled to produce high-resolution output.

2.1.1 Edge Guidance. Tables Given LR input $I_{\{LR\}}$, edge priors are computed via gradient magnitude:

$$\mathrm{E} = \sqrt{}\{(I\ \{LR\} * Gx)\ 2 + (I\ \{LR\} * Gy)\ 2\}$$

where G_x and G_y are Sobel kernels. These are concatenated with learned features: ***F'=Conca*(*F,E*)**, reinforcing boundary information.

2.1.2 Error-Aware Refinement. An error map $\Delta = |\mathrm{I}_{\{\mathrm{HR}\}} - \{\hat{I}\}_{\{\mathrm{SR}\}}|$ re-weights feature adaptively:

$\mathrm{F}'' = \mathrm{F}' \odot (1 + \alpha \cdot \Delta)$ focusing computation on challenging edge transitions.

2.1.3 Multi-Component Loss. We optimize: $\mathcal{L}_{total}=\lambda_1\mathcal{L}_{\mathrm{L1}}+\lambda_2\mathcal{L}_{\mathrm{edge}}+\lambda_3\mathcal{L}_{\mathrm{perceptual}}$ where $\mathcal{L}_{\mathrm{L1}}$ensures pixel accuracy, $\mathcal{L}_{\mathrm{edge}}$ preserves structural boundaries, and $\mathcal{L}_{\mathrm{perceptual}}$ enhances visual realism [Zhang et al., 2018]. Ablation studies determined $\lambda_1 = 1.0$, $\lambda_2 = 0.6$, $\lambda_3 = 0.4$.

As illustrated in Figure 2, the architectural comparison clearly delineates the evolution from the baseline EDSR framework to our proposed EEDSR network. Figure 1(a) depicts the original EDSR structure, which relies solely on a sequence of residual blocks for feature extraction. In contrast, our EEDSR architecture, shown in Figure 1(b), introduces two critical modifications: (1) an edge guidance pathway that extracts and injects explicit edge information directly into the feature learning process, and (2) a multi-component optimization module that computes the composite loss L total during training. This visual comparison underscores that our enhancements are strategically integrated alongside the proven residual blocks, maintaining architectural efficiency while directly addressing the core limitations of edge preservation. Furthermore, the edge-guided branch operates in are reinforced at multiple depths of the network.

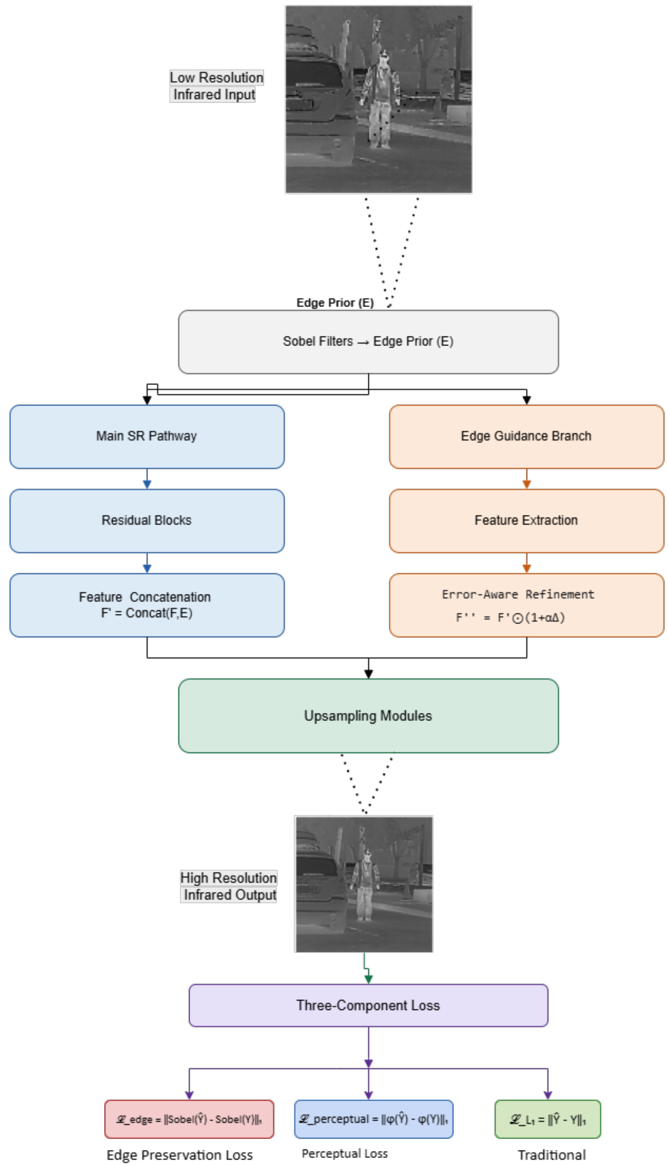

Figure 1: Overall workflow of the proposed EEDSR framework, showing the parallel processing of main super-resolution pathway and edge guidance branch, followed by feature fusion and multi-component optimization.

2.2 Implementation

EEDSR maintains EDSR's residual configuration (B=32, F=256) for fair comparison, trained with Adam (LR=10^{-4}, halved every 30 epochs) and data augmentation. This framework specifically targets thermal edge preservation while maintaining computational efficiency

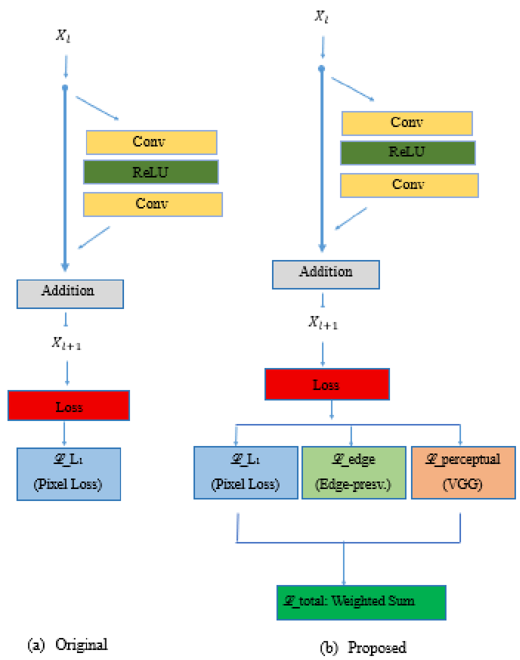

Figure 2: Architectural comparison illustrating (a) baseline EDSR framework and (b) our proposed EEDSR with integrated edge guidance pathway and multi-component optimization, maintaining identical core residual blocks while significantly enhancing boundary preservation capabilities.

3 Experiments

3.1 Datasets

We evaluate EEDSR on the M3FD benchmark and a real-world dataset captured with a CONOTECH Aquila thermal camera (640×512, 8–14 μm). M3FD provides controlled infrared-visible pairs; our real-world data introduces sensor drift, motion blur, and environmental noise for robustness testing.

3.2 Training Details

Our training methodology employs a systematic approach to ensure fair comparison and reproducible results. We extract infrared image patches of size 128×128 from low-resolution inputs, with corresponding high-resolution patches of size 512×512 serving as ground truth. To enhance model generalization and prevent overfitting, we implement comprehensive data augmentation including random horizontal flips and 90-degree rotations. All images undergo preprocessing through pixel value normalization to the [0,1] range, ensuring numerical stability during optimization.

We employ the Adam optimizer [Kingma and Ba, 2014] with parameters β_1=0.9, β_2=0.999, and ϵ =10^{-8}, which provides stable convergence characteristics for deep super-resolution tasks. The training utilizes a minibatch size of 8 across all models, balancing memory constraints with gradient estimation quality. The learning rate follows a scheduled decay strategy, initialized at 10^{-4} and halved every 30 epochs to facilitate fine-grained convergence in later training stages.

All networks are trained from scratch on the M3FD infrared dataset to ensure fair comparison. The baseline SRCNN and EDSR models employ traditional MSE and L1 losses respectively, while our proposed EEDSR implements a composite loss function combining L1 reconstruction loss with edge-aware and perceptual components. Through empirical analysis, we determined that this composite loss provides superior convergence behavior and more effective structural information preservation compared to traditional loss formulations for infrared imagery.

Implementation is conducted using the PyTorch framework on NVIDIA RTX 3090 GPUs. The training process requires approximately 9 hours for EEDSR, 10 hours for EDSR, and 8 hours for SRCNN to complete 100 epochs. Comprehensive evaluation of all trained models on the test set necessitates approximately 6 hours to generate detailed metrics and error maps for comparative analysis.

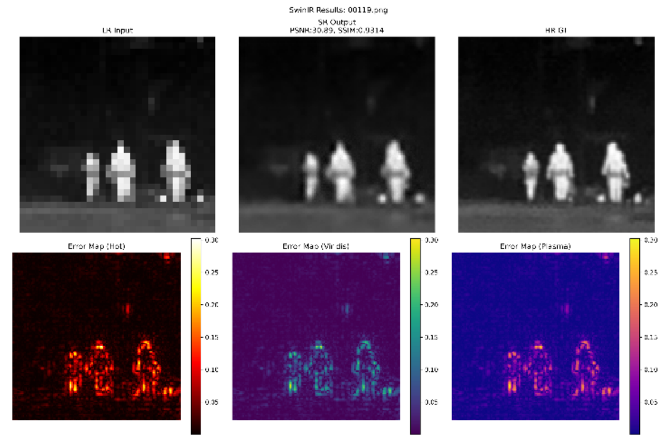

Figure 3: SwinIR performance on the M3FD benchmark

3.3 Ablation Study and Component Analysis

To quantitatively validate the contribution of each component in our proposed EEDSR framework, we conduct a systematic ablation study on the M3FD validation set. This analysis progressively integrates loss components to isolate their individual and collective impacts on reconstruction quality.

Table 1 shows component contributions on M3FD validation. Adding edge loss to EDSR+L1 boosts SSIM by 0.0195; adding perceptual loss further improves PSNR by 0.68 dB. Full EEDSR achieves optimal balance (37.37 dB PSNR, 0.9413 SSIM) with sharp edges and realistic textures.

3.4 Comparison with Transformer Based Methods

To address the need for contemporary benchmarking, we evaluated SwinIR [Liang et al., 2021], a state-of-the-art transformer-based super-resolution model. As shown in Table 2, SwinIR achieves competitive performance on the M3FD benchmark (PSNR: 30.89 dB, SSIM: 0.9314) due to its ability to model long-range dependencies.

Table 1: Ablation study of loss components on M3FD validation set

Model Configuration	PSNR (dB)	SSIM	Visual Quality Assessment
EDSR + L1 only	32.29	0.8529	Good pixel accuracy, blurred edges
+ Edge Loss	35.84	0.8724	Improved structural preservation
+ Perceptual Loss	36.52	0.8937	Enhanced texture realism
EEDSR (Full)	**37.37**	**0.9413**	**Optimal balance, sharp edges**

However, visual inspection reveals critical limitations as Figure 3 demonstrates that SwinIR introduces unnatural sharpening artifacts and fails to preserve thermal consistency along edges, particularly in homogeneous regions. This artifact stems from SwinIR's design for RGB imagery where sharp transitions are desirable, but in thermal data, they disrupt the smooth heat gradients essential for accurate temperature interpretation.

The error maps reveal high reconstruction errors along structural boundaries (head, shoulders) where SwinIR fails to recover correct shapes from heavily degraded inputs. This performance drop occurs because SwinIR, trained primarily on bicubic downsampled RGB data, lacks mechanisms to handle the complex, unknown degradations present in real thermal imagery (compression artifacts, sensor noise, motion blur, and gamma distortion). In contrast, our EEDSR maintains robust performance across both synthetic and real-world scenarios, demonstrating superior adaptation to thermal domain characteristics through explicit edge guidance and multi-component optimization.

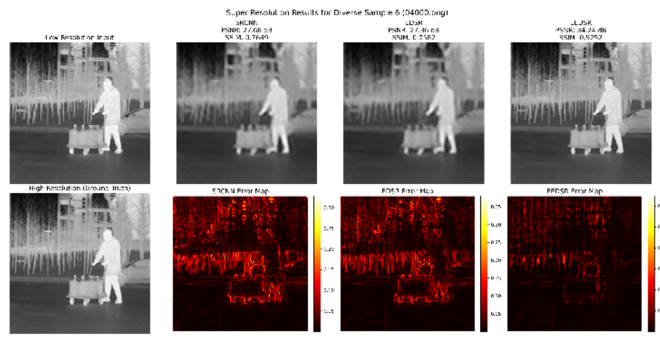

Figure 4: Comparative error maps on M3FD dataset showing reconstruction discrepancies for SRCNN, EDSR, and proposed EEDSR across diverse thermal scenes including urban environments and pedestrian scenarios.

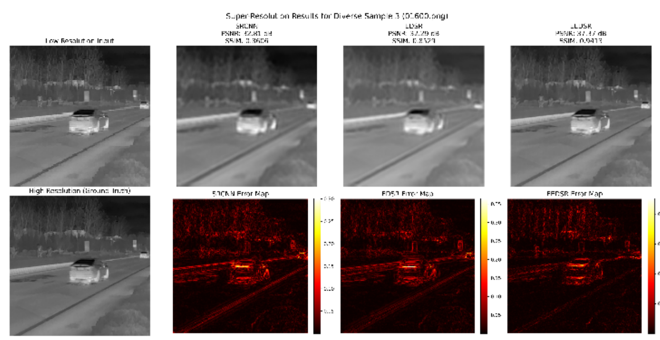

Figure 5: Super-resolution visual comparisons on M3FD dataset: vehicle and road sample with corresponding PSNR/SSIM metrics for SRCNN, EDSR, and proposed EEDSR.

3.5 Evaluation on M3FD Benchmark Results

The quantitative and qualitative evaluation on the M3FD benchmark demonstrates the superior performance of our proposed EEDSR model compared to established baseline methods. Visual error maps in provide critical insights into the limitations of existing super-resolution approaches when applied to infrared imagery.

SRCNN exhibits widespread reconstruction errors across both edge and non-edge regions—especially around building contours, vehicle boundaries, and human silhouettes—due to its shallow architecture, while EDSR reduces errors in smooth thermal areas through deeper residual learning; however, both methods still show elevated errors at sharp edges, fine structures, and high thermal-gradient regions that are crucial for infrared object detection and scene understanding.

Building upon these observations, Figure 4 illustrates the super-resolution results for two representative infrared test samples, accompanied by their corresponding quantitative metrics. For the vehicle sample, EEDSR achieves a PSNR of **37.37 dB** and SSIM of **0.9413**, substantially outperforming SRCNN (32.81 dB, 0.8606) and EDSR (32.29 dB, 0.8529). The accompanying error maps confirm this improvement, with EEDSR exhibiting significantly lower residual intensity across edge regions and object boundaries, indicating enhanced texture preservation and reduced reconstruction error.

Similarly, for the pedestrian scenario in Figure 5EEDSR attains **34.24 dB** PSNR and **0.9252** SSIM, again surpassing both SRCNN (27.68 dB, 0.7649) and EDSR (27.36 dB, 0.7582). Visually, EEDSR outputs retain sharper contours and clearer object details, particularly around edges and small-scale structures that appear blurred in baseline models.

These results collectively demonstrate that EEDSR effectively captures fine spatial details and restores high-frequency information with superior PSNR and SSIM scores. Overall, it achieves an optimal balance between perceptual quality and accuracy, confirming strong generalization for infrared super-resolution.

3.6 Real-World dataset Imagery

EEDSR maintains robust performance on real data (30.47 dB PSNR, 0.8810 SSIM) with sharp boundaries and thermal consistency. SwinIR shows substantial degradation (25.60 dB PSNR, 0.7670 SSIM), failing to recover structural details.

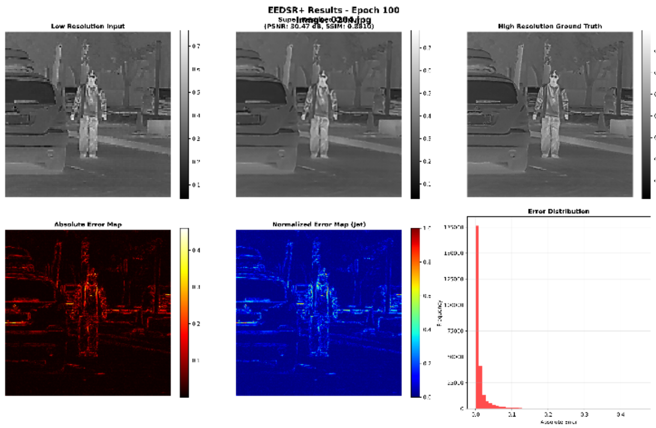

Figure 6: Real-world infrared super-resolution results: (a) Low-resolution input, (b) EEDSR reconstruction, (c) Ground truth reference, and (d) Error map visualization demonstrating minimal reconstruction discrepancies.

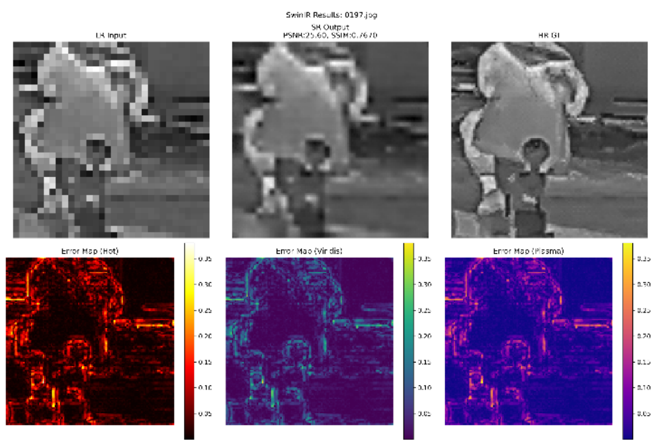

Figure 7: SwinIR's performance on real-world thermal image

As shown in Figure 6 EEDSR demonstrates excellent reconstruction performance on real-world captured data, exhibiting sharper structural boundaries, clearer edge definition, and more consistent intensity transitions. As compared to More critically, SwinIR's performance degrades significantly on real-world thermal data (PSNR: 25.60 dB, SSIM: 0.7670), as shown in Figure 7. While in EEDSR the textures appear naturally enhanced without introducing artificial artifacts, indicating effective feature representation learning under practical imaging conditions. Quantitatively, the model achieved a **PSNR of 30.47** and **SSIM of 0.8810**, reflecting high perceptual quality and structural similarity. The normalized and absolute error maps confirm that most discrepancies occur only in fine-detail regions, while overall reconstruction remains visually faithful.

The real-world evaluation highlights EEDSR's robustness against motion blur and sensor temperature drift, effectively reconstructing sharp contours and maintaining thermal consistency. It also excels in low-contrast scenes, recovering meaningful details where traditional methods often fail.

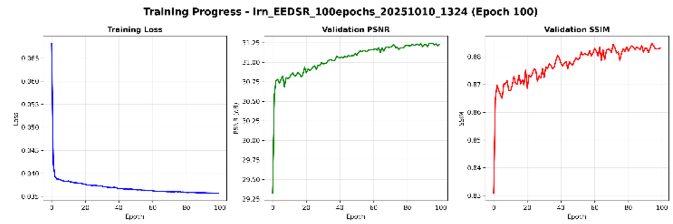

Figure 8: Training progression of EEDSR over 100 epochs: (left) training loss convergence, (middle) validation PSNR improvement, (right) validation SSIM enhancement, demonstrating stable optimization and effective learning.

The training dynamics, illustrated in Figure 8, show rapid convergence and continuous improvement in validation metrics, confirming the effectiveness of our edge- and perceptual-aware loss functions. We employ carefully tuned weighting parameters ($\lambda_1 = 1.0$, $\lambda_2 = 0.6$, $\lambda_3 = 0.4$) determined through extensive ablation studies to balance pixel accuracy, edge preservation, and perceptual quality.

Table 2 summarizes the comprehensive evaluation across both benchmark and real-world datasets. On the professionally captured M3FD dataset, EEDSR significantly outperforms baseline models, reflecting enhanced capability to reconstruct fine structural boundaries and retain high-frequency details. The real-world validation, despite introducing higher uncertainty from handheld imaging constraints, confirms EEDSR maintains stability and resilience under practical thermal imaging conditions.

4 Conclusion

This work presents EEDSR, an edge-enhanced super-resolution framework developed to meet the structural demands of infrared imaging. The model strengthens a residual backbone with explicit edge guidance and a carefully balanced combination of reconstruction, edge-aware, and perceptual objectives. Comprehensive evaluations and detailed error analyses show that EEDSR preserves fine structural boundaries with far greater reliability than existing approaches and maintains stable performance on real thermal captures, even in the presence of noise and environmental variation. These gains come from targeted optimization rather than an increase in model size, and they demonstrate the value of integrating domain-specific structural cues into infrared super-resolution. Future research will extend the framework to video, where temporal consistency becomes essential, and will adapt its edge-aware learning principles to related thermal vision tasks such as segmentation and object detection.

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Table 2: Comprehensive quantitative evaluation across benchmark and real-world datasets

Dataset	Model	PSNR (dB)	SSIM	Visual Quality Assessment
M3FD (Vehicle)	SRCNN	32.81	0.8606	Baseline
	EDSR	32.29	0.8529	Improved smooth regions
	SwinIR	30.89	0.9314	High SSIM but edge artifacts
	EEDSR	37.37	0.9413	Best overall balance
M3FD (Pedestrian)	SRCNN	27.68	0.7649	Very sharp error on edges
	EDSR	27.36	0.7582	Limited edge preservation
	SwinIR	30.89	0.9314	High SSIM but edge artifacts
	EEDSR	34.24	0.9252	Sharp contours, clear details
Real-World	SwinIR	25.60	0.7670	Poor generalization
	EEDSR	30.47	0.8810	Robust to real degradation

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